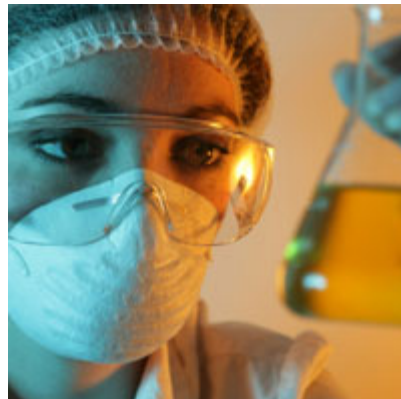


# Accelerating the Drug Development Lifecycle



# Goals

- ☐ Reduce effort & time for individual projects (cycle time)
- ☐ Reduce risks (associated with change & variability)
- ☐ Improve extendibility
- ☐ Improve adaptability
- ☐ Improve resiliency
- ☐ Improve flexibility
- ☐ Improve knowledge sharing (globally)
- ☐ Improve collaboration w/in & w/out R&D

## Use Cases

### ❑ Reduce effort & time for individual projects (cycle time)

- Shift from manual to automated workflows will reduce handoffs and transitions between staff
- Decisions are accelerated by having information available in a more timely fashion
- Shift from a document-centric process to one that is data centric to reduce overall effort and costs for a project.
- Real time monitoring of processes, projects and programs will reduce the need for status reports and meetings for program managers and management.
- Workload balancing along with identification of latency in process flows will allow management to optimize cycle time.

## Use Cases

### ❑ Reduce risks (associated with change & variability)

- Changes to processes and/or SOPs can be made, reviewed and pushed out w/in hours or days.
- Anomalies in the process can be addressed in real-time rather than incurring extended wait times
- Implementing changes should not require extensive training since the process should drive the right behaviors and decisions (reflective of the changes)
- Knowledge is captured along the way, embedded in the process and made available to mitigate impact due to staff attrition and/or learning curves for process changes.
- Support electronic signature and enable security over processes and data throughout the lifecycle.

# Use Cases

## ❑ Improve Extensibility

- The same process framework and knowledge repository should be able to support product development within the current drug pipeline as well as future products (breadth).
- The process should be able to handle existing volume of projects in the generic drug development pipeline and scale to handle more project volume in the future (scale).
- New features can be easily added to the automation platform w/ little effort and w/o impacting existing projects already in progress (breadth).

# Use Cases

## ❑ Improve Adaptability

- Processes adapt to changing business conditions and/or unplanned events w/o IT intervention
- Business rules can be maintained and updated to address new conditions and/or improve decision support as part of continuous improvement w/o IT intervention
- The process flows can be easily adapted from handling generic drug development to address biogenerics and more complex pharma engineering development.
- The system can capture & share new types of knowledge as the drug development process evolves and expands w/o expensive changes to the platform and repository.

# Use Cases

## ❑ Improve Resiliency

- As anomalies arise, the processes will not simply fail if an unplanned condition is encountered.
- If errors occur, the system will attempt to self-heal itself for simple errors and route more complex errors to knowledge engineers and/or support rather.
- The system should be fully auditable with end-to-end traceability where issues and/or deliverables can be reproduced at any point w/in the process flow.
- A fail forward process should be built into the platform to ensure the system continuously “learns” and improves.

# Use Cases

## ❑ Improve Flexibility

- New functionality should require decreasing amounts of effort over time as the platform matures (suggests reuse is a core competency).
- Existing component processes should be able to be leveraged for secondary and tertiary processes which are used as inputs to (or outputs from) the core drug development processes.
- Process behavior (directional flow) should be easily modified w/o significant effort (initially light IT involvement, eventually no IT involvement).



# Use Cases

## ❑ Improve knowledge sharing (globally)

- Data and knowledge is captured at every point in the lifecycle for projects
- Knowledge is made available for resources to consume through publish or subscription services
- Ability to find knowledge thru explicit & implicit search
- Knowledge is shared according to the 5 Rights:
  - Right Information
  - Right Format
  - Right Timing
  - Right made
  - Right amount

# Use Cases

## ❑ **Improve collaboration w/in & w/out R&D**

- Information from forums, wikis, blogs, discussions and other social networking tools will be captured and aligned with process deliverables, projects and programs.
- Individuals (w/in & w/o R&D) will be able to collaborate on projects, programs across the entire development lifecycle
- Individuals will be able to search across all types of collaborative content and find related knowledge associated with projects and deliverables.
- Teams will be able to share lessons learned for projects to promote continuous learning

# Key Platform Disciplines

## ❑ Business Process Automation (BPMS)

- Automated workflow & services
- On Demand monitoring, load leveling & governance

## ❑ Business Rules Automation (BRMS)

- Automated decision making
- Automated intelligence

## ❑ Configuration Management (ECM)

- Dynamic modeling (products, batch records, BoMs)

## ❑ Knowledge Management (KM)

- Collaboration (project, team, program, R&D)
- Document Management
- Learning & Knowledge Sharing

## Opportunities for Automation

- ❑ Individually, each of these automation investments would provide enterprise value to the company.
- ❑ Collectively, they will provide R&D with an agile, adaptable and scalable platform to achieve the current set of goals (prior section) as well as future business goals.
- ❑ The following slides provide a phased approach for implementing the automation investments along with a roadmap of capabilities that address the desired outcomes along the way.
- ❑ A more detailed overview of the architecture can be found in Appendix A along with an overview for each of the automation tools.

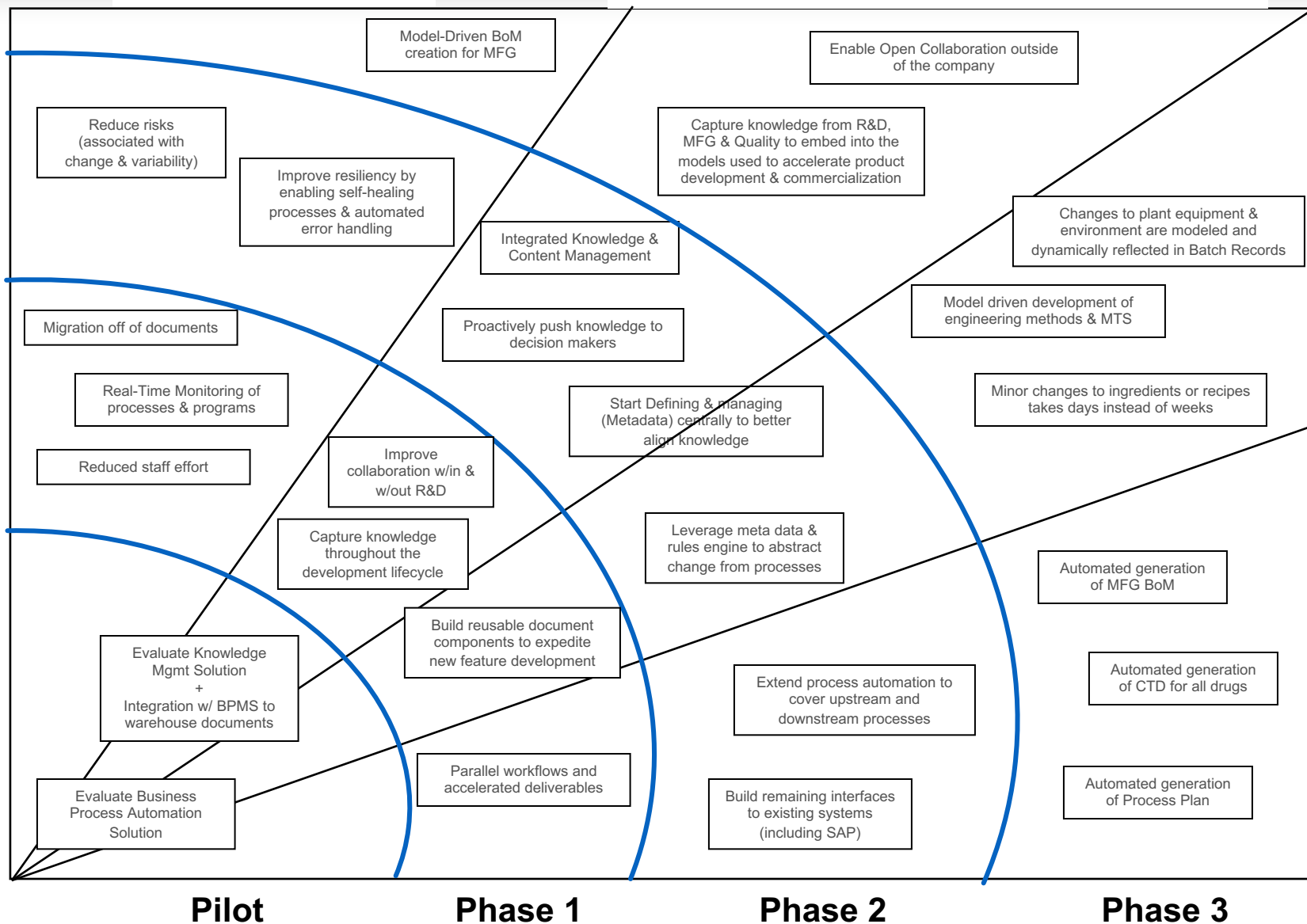
# Automation Roadmap

Efficiency

Collaboration

Adaptability

Speed



# Phased Approach

## ❑ Pilot

- Agree upon a representative scope & success criteria to evaluate the pilot
- Engage the vendor & their staff to build the scoped-out solution with input from selected team members (no IT development staff involved in the pilot)
- Recommend we ask the vendor to build the pilot solution on their own environment and provide access to the corporate staff participating in the pilot (min. costs & setup time)
- Decision makers to evaluate results (go/no go decision)

# Phased Approach

## ❑ Phase 1

- Migrate from a document-centric to a data centric solution
- Define a data model
- Streamline the processes (simplify & run in parallel)
- Implement a BPMS solution for Generic Drug Dev.
- Implement a dashboard for real-time monitoring
- Automate flow of deliverables into CTD Publishing Tool
- Automate MTS creation
- Implement an ECM solution to capture documents & support knowledge mgmt (see Appendix D)
- Enable Collaboration and align content with projects
- Implement integration points (see Appendix F)

# Phased Approach

## ❑ Phase 2

- Enhance process automation with intelligence
- Implement a rules engine (BRMS) (see Appendix C)
- Centralize rules management
- Implement self-healing & resiliency in the system
- Implement meta data management to align data & content
- Build proactive services to improve knowledge (5 Rights)
- Continue automating integration points (see Appendix F)
- Expand process automation to all drugs in all plants
- Automate partial feed of data into EBS build



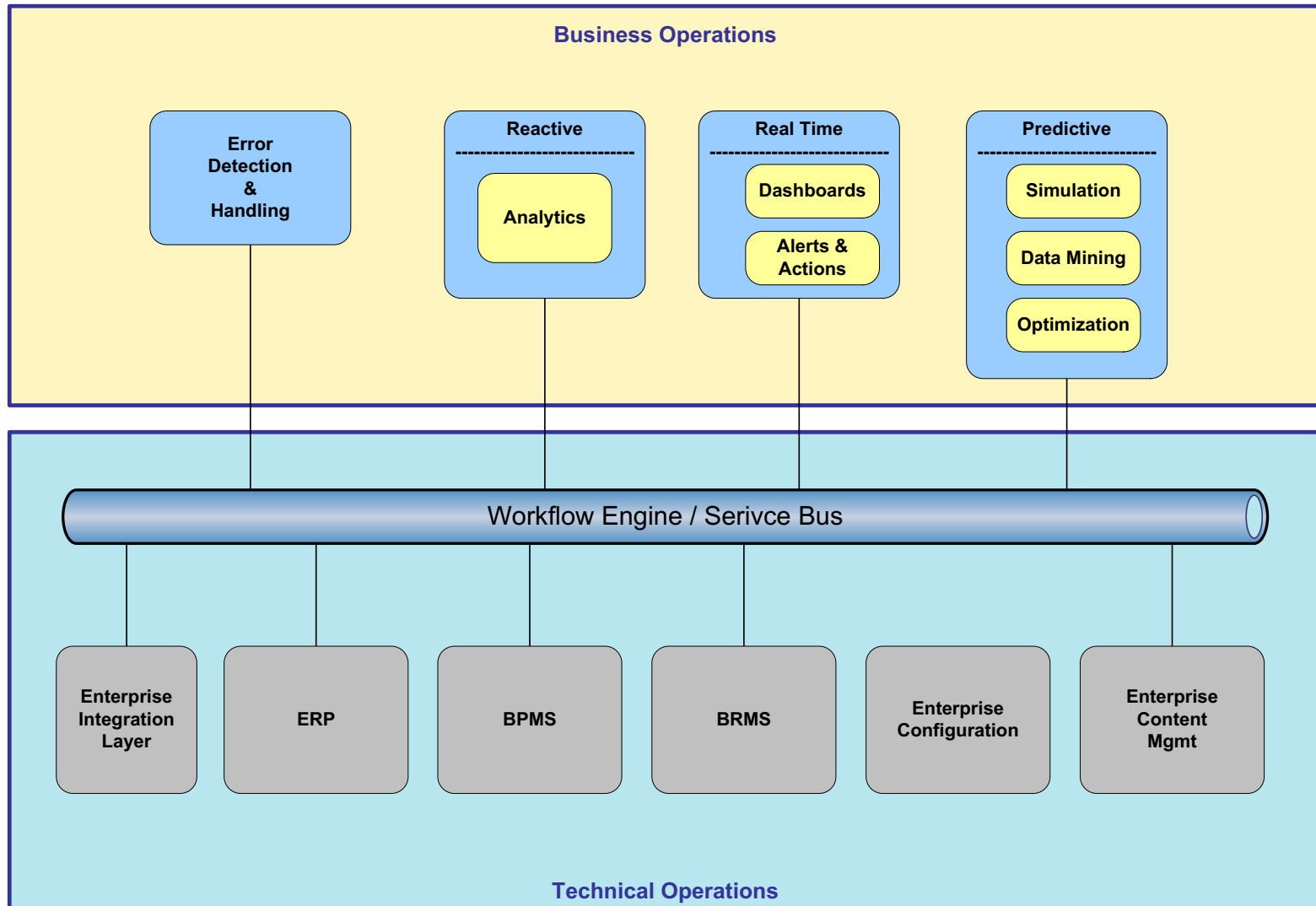
# Phased Approach

## ❑ Phase 3

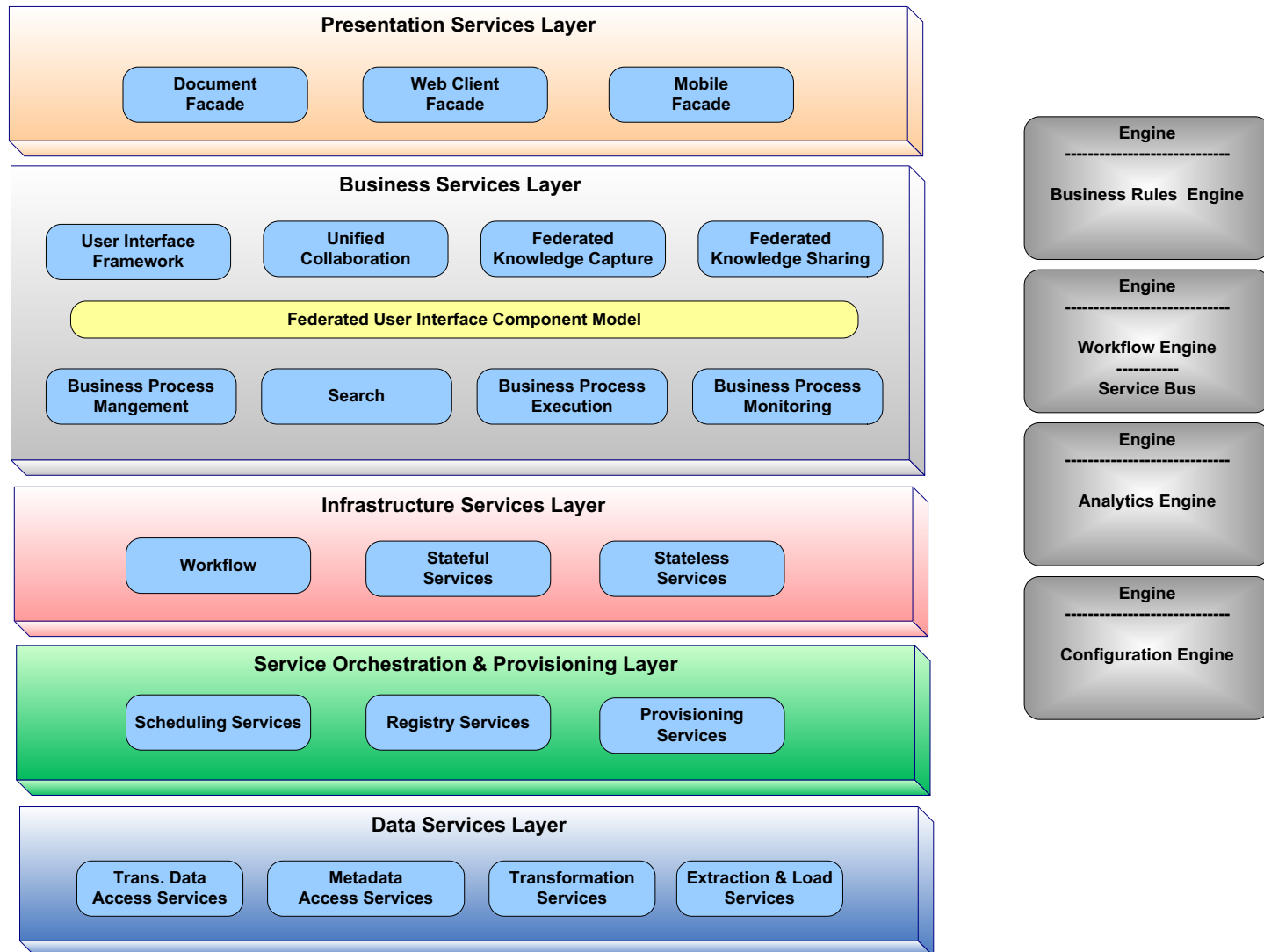
- Migrate from data-centric to model-driven
- Implement a configuration tool (see Appendix E)
- Model products, processes, plant, and equipment
- Implement an on demand solution that reduces cycle time for commercialization of product changes and new products introductions by an order of magnitude
- Automate the creation of the following artifacts:
  - Engineering Bill of Materials (eBoM)
  - Manufacturing Bill of Materials (mBoM)
  - Process Plan (Batch Record)

## Architectural Views

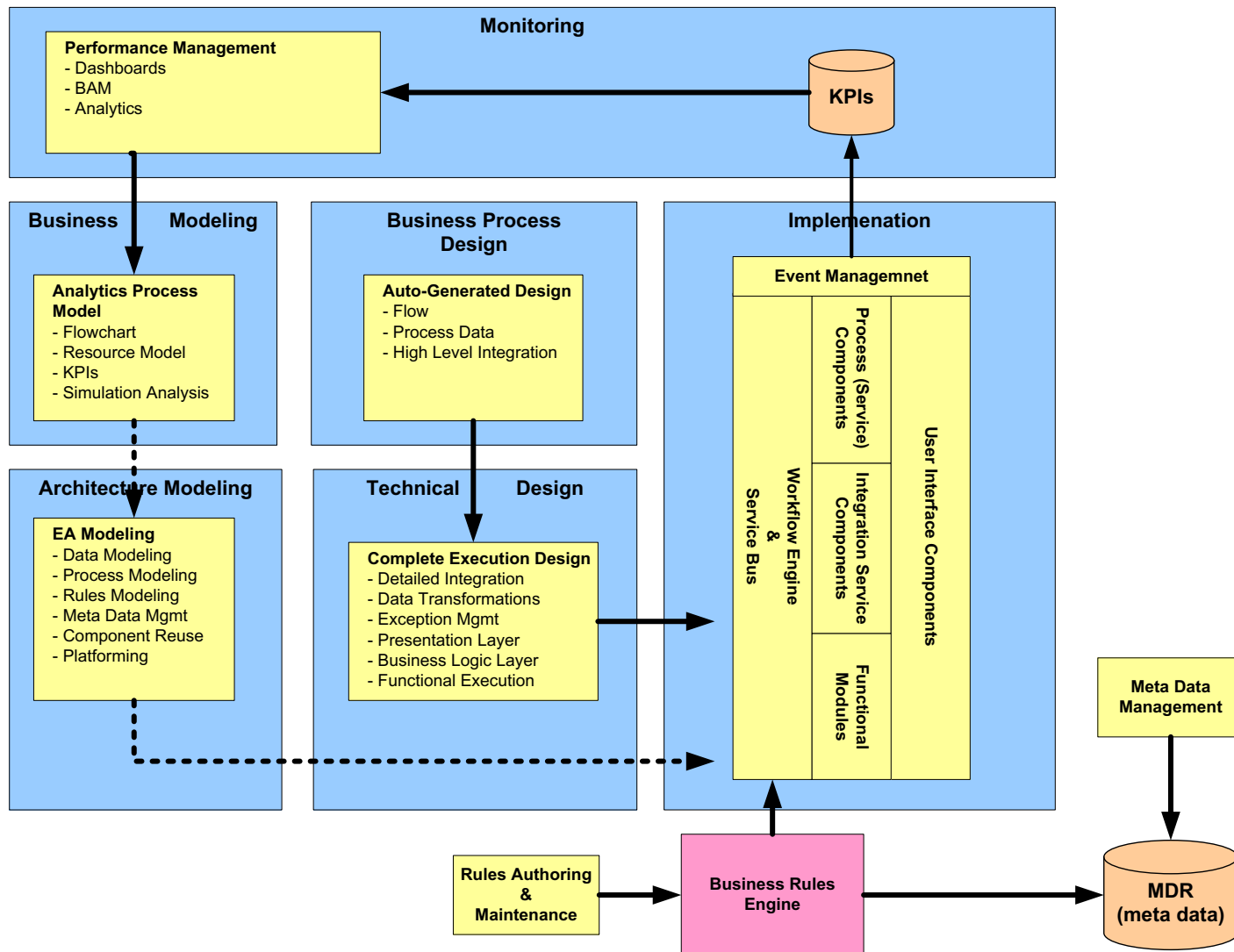
# Architecture View (Operational Services)



# Architecture View (Engines & Services)

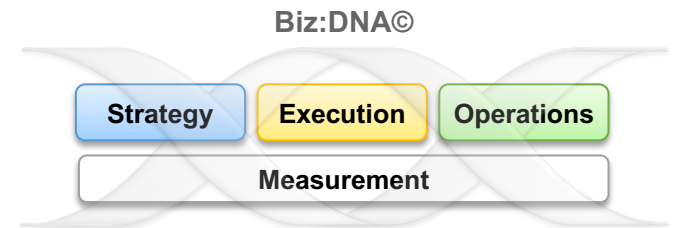


# Architecture View (Managing Change)



# Thank You

Measure what matters  
Create and deliver the right value  
Build and nurture a winning culture  
Proactively manage change & complexity



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